Evaluate yourself : 1

Calculate the molar conductance of 0.01M aqueous KCl solution at 25°C . The specific conductance of KC*l* at 25°C is 14.114×10^{-2} Sm^{-1} .

9.1.2 Equivalent conductance (Λ)

Equivalent conductance is defined as the conductance of 'V' m^3 of electrolytic solution containing one gram equivalent of electrolyte in a conductivity cell in which the electrodes are one metre apart.

The relation between the equivalent conductance and the specific conductance is given below.

$$\Lambda = \frac{\kappa\ (Sm^{-1}) \times 10^{-3}\ (\text{gram equivalent})^{-1}\ m^3}{N} \qquad .....(9..9)$$

Where κ the specific conductance and N is the concentration of the electrolytic solution expressed in normality.

Evaluate yourself : 2

The resistance of 0.15N solution of an electrolyte is 50 Ω. The specific conductance of the solution is 2.4 Sm^{-1}. The resistance of 0.5 N solution of the same electrolyte measured using the same conductivity cell is 480 Ω. Find the equivalent conductivity of 0.5 N solution of the electrolyte.

Given that

$R_1 = 50\ \Omega$ $\qquad$ $R_2 = 480\ \Omega$

$\kappa_1 = 2.4\ Sm^{-1}$ $\qquad$ $\kappa_2 = ?$

$N_1 = 0.15\ N$ $\qquad$ $N_2 = 0.5\ N$

$$\Lambda = \frac{\kappa_2\ (Sm^{-1}) \times 10^{-3}\ (\text{gram equivalent})^{-1}\ m^3}{N}$$

$$= \frac{0.25 \times 10^{-3}\ S\ (\text{gram equivalent})^{-1}\ m^2}{0.5}$$

$$\Lambda = 5 \times 10^{-4}\ Sm^2\ \text{gram equivalent}^{-1}$$

we know that

$$\kappa = \frac{\text{Cell constant}}{R}$$

$$\therefore \frac{\kappa_2}{\kappa_1} = \frac{R_1}{R_2}$$

$$\kappa_2 = \kappa_1 \times \frac{R_1}{R_2}$$

$$= 2.4\ Sm^{-1} \times \frac{50\ \Omega}{480\ \Omega}$$

$$= 0.25\ Sm^{-1}$$

9.1.3 Factors affecting electrolytic conductance

If the interionic attraction between the oppositely charged ions of solutes increases, the conductance will decrease.

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